



Pharm*Achieve*

ACUTE OTITIS MEDIA (AOM)

Qualifying Exam Preparatory Course

COMPETENCIES COVERED IN THIS PRESENTATION...

**Providing Care:
Clinical Care**

**Providing Care:
Distribution**

**Knowledge
& Expertise**

**Communication
& Collaboration**

**Leadership &
Stewardship**

Professionalism

LEGEND

AOM	Acute Otitis Media
BID	Twice Daily
BP	Blood Pressure
ED	Emergency Department
ET	Eustachian tube
HD	High-Dose
HR	Heart Rate
IM	Intramuscular
IV	Intravenous
MEE	Middle Ear Effusion
O/E	On Examination
OME	Otitis Media with Effusion
PO	By Mouth
RSV	Respiratory Syncytial Virus
SD	Standard-Dose
TID	Three Times Daily
URTI	Upper Respiratory Tract Infection

DEFINITIONS

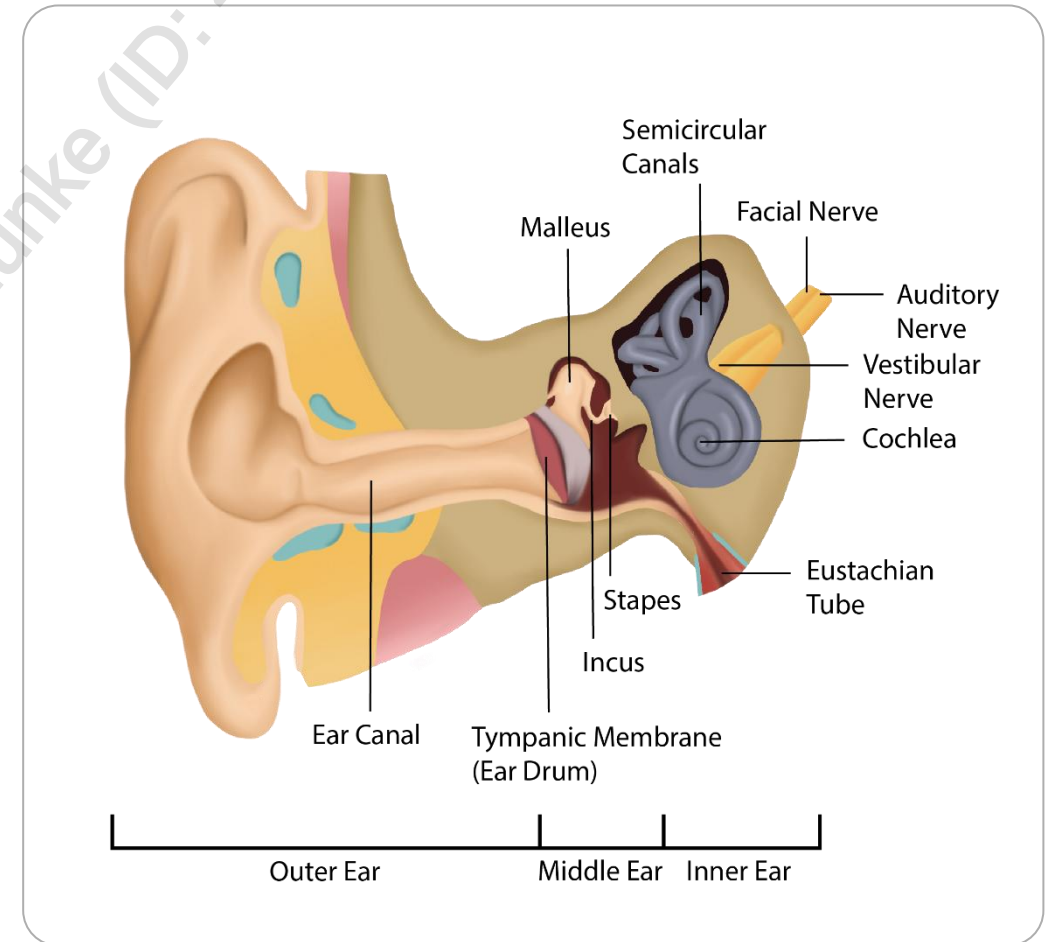
Most common in children and is often preceded by a viral or bacterial upper respiratory tract infection

Acute Otitis Media (AOM)

- Presence of inflammation in the middle ear accompanied by the rapid onset of **signs and symptoms of an ear infection**

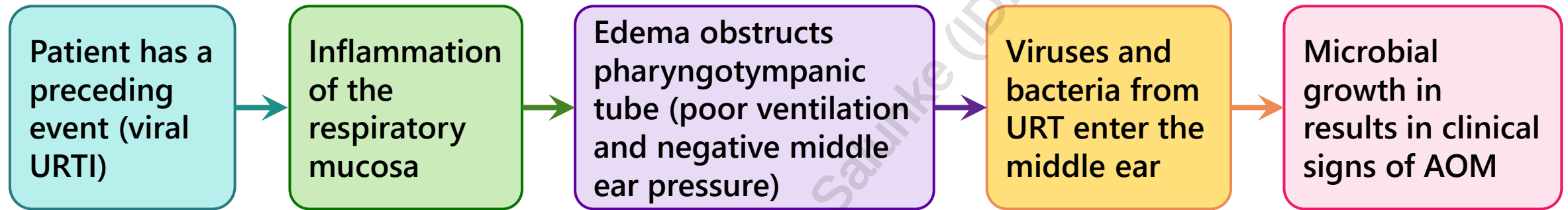
Otitis Media with Effusion (OME)

- Presence of fluid in the middle ear **without signs and symptoms of an ear infection**



PATHOPHYSIOLOGY OF AOM

FYI



Inflammation and fluid accumulation in the middle ear:

- The middle ear is the location of the hearing apparatus
- It is separated from the outer ear canal by the tympanic membrane (ear drum)
- Normally, the eustachian tube protects the middle ear from nasopharyngeal secretions and provides drainage from such secretions

ETIOLOGY AND EPIDEMIOLOGY

70% of bacterial infections occur due to the following pathogens	Up to 50% of AOM cases are due to viral causes listed (rarely fungal) and resolve without antibiotic therapy
• <i>S. pneumoniae</i> (25-50%) - 20% spontaneous resolution • <i>H. influenzae</i> (15-30%) - 50% spontaneous resolution • <i>M. catarrhalis</i> (3-20%) - 75% spontaneous resolution • <i>H. influenzae</i> and <i>M. catarrhalis</i> are likely to resolve spontaneously	• Respiratory Syncytial Virus (RSV) • Parainfluenza viruses • Influenza virus • Coronavirus • Rhinovirus • Adenovirus

Epidemiology

- Peak incidence between **6 - 18 months** (<2 years old)
- Common in **6 months – 3 years of age**: affects about 70% of children
- Uncommon after 8 years of age
- Recurrent AOM affects 15-30 % of otitis-prone children

Likely bacterial if:

- ☐ Bulging tympanic membrane that is hemorrhagic or yellow
- ☐ Perforation of the tympanic membrane with purulent discharge (otorrhea)

CLINICAL PRESENTATION

- The 3 most common symptoms include:
 1. Otolgia (ear pain)
 2. Irritability
 3. Fever (absent in 50% of cases)
- Patients may also experience nasal congestion, coughing, loss of appetite, vomiting and otorrhea

Severity of illness:

Mild AOM	Moderate - Severe AOM
• Absence of pain or mild pain • Alert, responsive, able to sleep • Oral temperature <39°C without antipyretic • <48 hours of symptoms	• Moderate to severe otalgia • Tugging on the ear, inability to sleep AND/OR • Oral temperature ≥39°C AND/OR • ≥48 hours of symptoms

The most common complication of AOM is acute mastoiditis

DIAGNOSIS

FYI

- AOM is likely if the following are present:
 - Acute onset (e.g. otalgia)
 - Middle-Ear Effusion (MEE)
 - Middle-Ear inflammation
- Methods for diagnosis: Visualizing the tympanic membrane with an otoscope
 - Bulging
 - Redness/opacity
 - Immobility
 - Perforation (high-sensitivity for bacterial cause, warranting antibiotics)
- Differential diagnoses include otitis media with effusion (OME, can last days to weeks after infection resolved), acute otitis externa, impacted earwax, chronic otitis media
- Recurrent AOM: ≥ 3 episodes in 6 months or ≥ 4 episodes in 12 months

RISK FACTORS

- Age: Children peak between 6 months – 3 years of age
- Childcare attendance or presence of siblings in the home
- Crowded household
- Pacifier use
- Second-hand smoke exposure
- Orofacial abnormalities (e.g. cleft palate)
- Recent episode of influenza/URTI
- Indigenous/Inuit/ children of First Nations (due to access to care/systemic inequity)
- Family history of recurrent otitis media
- Low socioeconomic status
- Immunocompromised patients
- Feeding while lying down, shorter duration of breastfeeding

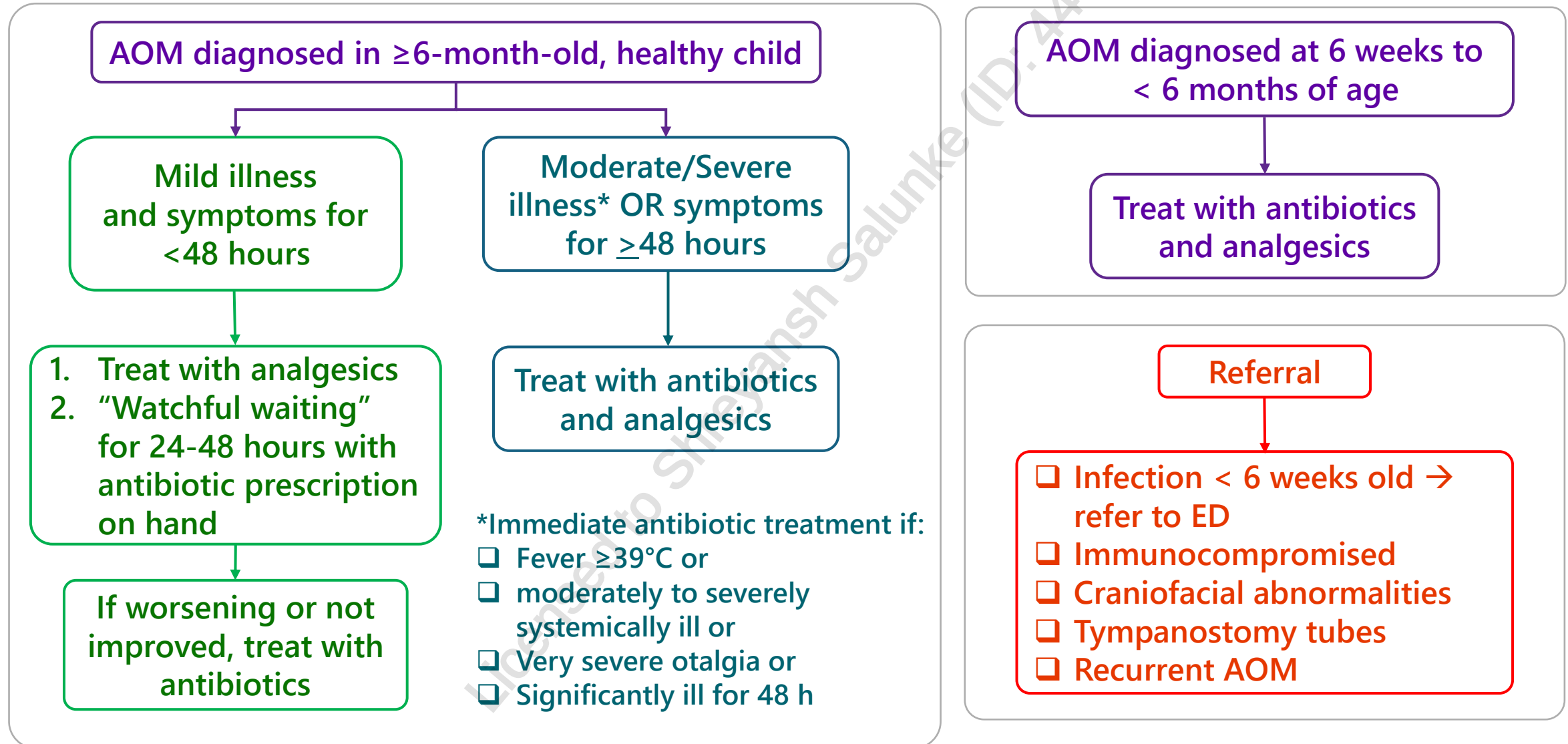
GOALS OF THERAPY

- ✓ Resolve signs and symptoms
- ✓ Eradicate infection and sterilize the middle ear
- ✓ Prevent complications
- ✓ Avoid inappropriate use of antibiotics

VACCINATION

- The annual **influenza vaccine** is recommended for all individuals ≥ 6 months, especially those with chronic medical conditions*:
 - *Renal disease, cardiac or pulmonary disorders (e.g. cystic fibrosis, asthma), metabolic disorders (e.g. diabetes), cancer, immunocompromised status, anemia, or neurodevelopmental conditions
- The **pneumococcal conjugate vaccine** is recommended in children as young as 2 months
- Non-typable *H. influenzae* strains cause the majority of AOM cases, limiting the effectiveness of the *Haemophilus influenzae* type b (Hib) vaccine

AOM TREATMENT ALGORITHM



WATCHFUL WAITING

When should watchful waiting be considered?

- During the first 24 - 48 hours in children ≥ 6 months of age with the following:
 - ☐ Mild illness
 - ☐ Uncomplicated AOM, temperature $< 39^{\circ}\text{C}$
 - ☐ No serious comorbidities (e.g. immunodeficiency)
 - ☐ Guardian can recognize when illness is worsening and return for follow-up

Conditions for watchful waiting

- Option is indicated for patients whose infection is expected to resolve spontaneously
- Caregiver should have ready access to clinicians and should return if the child does not improve or worsens anytime within 48 hours
- If antibiotic prescription is issued, the parent is asked not to fill the prescription unless symptoms persist or worsen over 24-48 hours

TREATMENT OVERVIEW

- ❑ Amoxicillin is considered first-line therapy for acute otitis media (AOM), as infection is predominantly caused by *S pneumoniae*
- ❑ *M catarrhalis* and *H influenzae* can be resistant to amoxicillin due to their ability to produce beta-lactamases, but infection caused by those pathogens is less common and more likely to resolve spontaneously
- ❑ In patients with a history of penicillin or amoxicillin hypersensitivity, use second or third-generation cephalosporins UNLESS the previous reaction was life-threatening (e.g. hypotension, bronchospasms)
 - ❑ Alternative agents for patients with life-threatening hypersensitivity reactions: macrolides (e.g. azithromycin or clarithromycin) or clindamycin → less efficacy against *S pneumoniae* and *H influenzae*

Patient Counselling:

- ❑ Symptoms should begin to improve within 24 hours and typically resolve within 2 to 3 days after starting antimicrobial therapy
- ❑ Persisting or worsening symptoms require additional assessment for initial treatment failure

FIRST-LINE ANTIBIOTICS

Duration of beta-lactam therapy:

- ☐ ≥ 2 years (uncomplicated): 5 days
- ☐ < 2 years AND/OR relapse/failure/perforated TM: 10 days

Drug	Dose	Comments
First-line therapy if no penicillin allergy:		
SD amoxicillin	45-60mg/kg/day in 3 divided doses as a capsule or suspension	• Only use in children with no daycare exposure or no antibiotic use in 90 days • SD not common due to risk of resistant <i>S. pneumoniae</i>
HD amoxicillin	75-90mg/kg/day in 2-3 divided doses as a capsule or suspension	• With risk factors for <i>S. pneumoniae</i> resistance [e.g. recent antibiotic use (< 3 months), daycare attendance]
First line therapy in patients with non-life-threatening penicillin allergy:		
Cefuroxime axetil	30 mg/kg/day in 2-3 divided doses as a capsule or suspension	
Ceftriaxone	50mg/kg IM/IV daily for 3 days	• In patients unable to tolerate oral route

INITIAL TREATMENT FAILURE (NO IMPROVEMENT AFTER 2-3 DAYS)

Drug	Dose	Comments
Amoxicillin-clavulanate	- If weight of child ≤ 35 kg: 45-60 mg/kg/day in 3 divided doses x 10 days - If weight of child >35 kg: 500/125 mg PO TID x 10 days	- Preferred suspension in Canada is the 7:1 (amoxicillin:clavulanate) formulation - More diarrhea than amoxicillin (due to clavulanate)
Ceftriaxone	50 mg/kg/day x 3 days	- IV or IM injections - Reserved for patients who cannot tolerate oral antibiotics or have failed amoxicillin-clavulanate

OTITIS-CONJUNCTIVITIS SYNDROME

FYI

- Also known as AOM with purulent conjunctivitis
- Common pathogens: *H. influenzae* and *M. catarrhalis*
- Symptoms include otalgia, irritability, red watery eyes and purulent discharge
- Treated with beta-lactamase inhibitors
 - ❑ Amoxicillin-clavulanate or second-generation cephalosporins (e.g. cefuroxime axetil)

MONITORING PARAMETERS

Efficacy Endpoints	Degree of change	Timeframe
Pain	Reduce	24 hours
	Eradicate (0/10)	48 - 72 hours
Fever	Back to normal (varies slightly based on route of measurement)	24 - 48 hours
Appetite/ behaviour	Typical appetite	24 - 72 hours

Safety Endpoints	Degree of change	Timeframe
Drug side effects e.g. nausea, vomiting, diarrhea	Minimize	Duration of therapy

TAKEAWAYS

- Acute otitis media is a common infection in pediatric patients
- Understanding which patients to treat immediately and which to employ “watchful waiting” is a key objective of management
- Analgesia is an important component of care in all patients, regardless of antimicrobial choice

Treatment:

- Ensure weight-based dosing is correct in all patients
- **Amoxicillin** is considered first-line therapy in patients without penicillin allergy, as infection is predominantly caused by *S. pneumoniae*
- **Cefuroxime-axetil or ceftriaxone** can be used alternatively as first-line therapy in patients with non-life-threatening penicillin allergy
- **Amoxicillin/clavulanate** is used in patients who failed initial amoxicillin therapy

Duration of therapy:

- ≥ 2 years (uncomplicated): 5 days
- < 2 years AND/OR relapse/failure/perforated TM: 10 days

CASE 1

SL is a 2-year-old girl who presents to her GP with fever, ear pain and decreased oral intake. SL had symptoms of a cough and cold 4 months ago and an episode of ear pain for which she received amoxicillin 30 mg/kg/day. She had recovered completely and was well until 3 days ago when her mother noted that she was tugging at her ear, crying more frequently and did not eat as much as usual. SL takes a children's chewable multivitamin 1 tab PO daily. She has no allergies to medications. She lives with her parents and 2 siblings and goes to daycare. Her father is an industrial chemical factory worker

Objective findings:

- BP=118/72 mmHg, HR=85 bpm
- Temperature= 39.5 C
- O/E: tympanic membrane opaque and bulging with limited mobility
- Weight=15 kg



CASE 1 - QUESTIONS

1. Which of the following increases SL's risk for acute otitis media?
 - a) Gender
 - b) Exposure to industrial chemicals
 - c) Daycare attendance
 - d) Recent antibiotic exposure
2. What therapeutic option would you suggest?
 - a) Watch and wait
 - b) Amoxicillin 450 mg po TID
 - c) Cefuroxime 280 mg po BID
 - d) Amoxicillin/clavulanate 300 mg po TID
3. What is the rationale for HD amoxicillin?
 - a) Overcome *H. influenzae* resistance
 - b) Ensure adequate penetration into middle ear
 - c) Allow for twice daily dosing
 - d) Overcome *S. pneumoniae* resistance
4. Which symptoms prompted you to treat SL as severe OM?
 - a) Otalgia severity
 - b) T=39.5°C
 - c) Recent URTI
 - d) Otalgia severity and T= 39.5°C

CASE 1 - QUESTIONS

5. SL is to receive amoxicillin 450mg TID for 10 days, but the pharmacy only has 100 mL bottles of the amoxicillin 125mg/5ml suspension available. Calculate the total dose in millilitres that the patient will receive per day.

- a) 27ml
- b) 54ml
- c) 30ml
- d) 62ml

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CHANGE LOG

July 2025

- Formatting changes made throughout; Legend updated
- Slide 6: Add note on indications of a bacterial cause
- Slide 7: changed > to $\geq$ 48 hours
- Slide 9: Minor additions to risk factors
- Slide 12: Major edits to the algorithm
- Slide 14: Added new slide on 'Treatment Overview'
- Slide 15: Minor change in cefuroxime dosage
- Slide 16: Added ceftriaxone
- Slide 17: Added cefuroxime as an example of a second-generation cephalosporin
- Slide 21: Edited options for question 4

October 2025

- Formatting changes made throughout
- Slide 9: Added risk factors
- Slide 11: Added examples of chronic medical conditions, removed statistics, and reworded *H. influenzae* bullet point
- Slide 12: Reworded 'AOM diagnosed at 6 weeks to < 6 months of age'
- Slide 19: Elaborated on takeaway slide

March 2026

- Slide 2: Updated NAPRA competencies
- Slide 18: Updated monitoring parameters
- Formatting changes made throughout